

CSS Animations & Transitions

Make Your Webpage Come ALIVE!



Beginner



Code Editor + Browser



Objective: Understand CSS transitions and create smooth hover effects



Keep coding — keep creating magic!



What Is Animation?

Animation makes things MOVE, CHANGE, or come ALIVE — guiding attention and making web experiences feel polished!



Swiggy Loader

Bike rides across screen while food is prepared



PhonePe Tick

Green tick appears after a successful payment



Google Doodle

Letters bounce or transform on special days



Diwali Diya

Flame flickers on festival websites



Flipkart Buttons

Glow or grow when you hover over them

Without animation: flat & jarring → With animation: alive & professional ✨

CSS — The Magic Wand

CSS (Cascading Style Sheets) controls colours, sizes, fonts, spacing — and YES, animations! A few lines of code tell the browser how to move, change colour, grow, shrink, or spin.

Two CSS Animation Tools

CSS Transition

 What: Smooth change between two states

 When: On hover, click, or focus

CSS @keyframes

 What: Continuous or complex movement

 When: Looping effects, loading spinners

★ Session focuses on CSS Transition. @keyframes comes in Session 2!

CSS Transition: The Slow-Change Tool

Real-Life Analogies



Traffic Light

Red → Yellow → Green — smooth, ordered transitions



Sunrise Sky

Black → purples → pinks → oranges — a slow beautiful blend



Flipkart Button

Colour fades from white to orange over a fraction of a second

The Transition Syntax

```
/* Basic syntax */  
transition: background-color 1s;  
  
/* Multiple properties */  
transition: background-color 1s, transform 0.5s;  
  
/* Transition ALL properties */  
transition: all 0.8s;
```

The 3 Parts of a CSS Transition 1 2 3 4

1

The Selector

```
.box { }
```

Targets which element gets the magic — a div, button, image, or any HTML element.

2

The Transition Property

```
transition: background-color 1s;
```

The magic line! What to change + how long. Goes in the NORMAL style, NOT in :hover.

3

The Hover Trigger

```
.box:hover { }
```

Fires when the user's mouse moves over the element. Like a motion-sensor light switch!

Without vs. With Transition

Without Transition

```
.box {  
  background-color: orange;  
}  
  
.box:hover {  
  background-color: green;  
}
```

 Color flips instantly — jarring and abrupt!

VS

With Transition

```
.box {  
  background-color: orange;  
  transition: background-color 1s;  
}  
  
.box:hover {  
  background-color: green;  
}
```

 Color slides smoothly over 1 second — natural!

 **Key Rule:** Put the **transition** property in the NORMAL style, not inside **:hover**!

Duration: How Fast or Slow?



0.2s–0.3s

Very Fast

Quick blink

✓ Micro-interactions

0.5s

Fast

Snappy & clean

✓ Button colour changes

1s

Medium

Comfortable

✓ Most hover effects

2s–3s

Slow

Dramatic sunrise

✓ Loading, special reveals

5s+

Very Slow

Dreamy

✓ Background colour shifts

Properties You Can Transition



```
background-color
```

Fill colour

orange → green



```
font-size
```

Text size

16px → 24px



```
transform: scale()
```

Grow / shrink

scale(1) → scale(1.3)



```
color
```

Text colour

black → white



```
border-radius
```

Corner shape

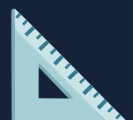
0px → 50%



```
transform: rotate()
```

Spin

0deg → 45deg



```
width / height
```

Element size

100px → 200px



```
opacity
```

Transparency

1 → 0.5



```
box-shadow
```

Drop shadow

none → 0 4px 20px ...



Hands-On Activity

MUST DO

Colour Change on Hover

Create a coloured box that smoothly changes background colour on hover. Use `transition: background-color 1s;` in `.box` style.

SHOULD DO

Try Three Speeds

Change duration to 0.3s, 1s, and 3s one at a time. Notice how each feels different. Which feels most natural?

STRETCH

Grow on Hover

Add `transform: scale(1.3);` in `.box:hover`. Transition both colour AND transform together with commas!

★ Starter Code

```
<style>
.box {
  width: 200px;
  height: 120px;
  background-color: orange;
  border-radius: 12px;
  /* 📍 ADD HERE */
  transition: bg-color 1s;
}

.box:hover {
  background-color: green;
  /* scale(1.3); */
}
</style>
```



Common Mistakes to Avoid



Transition inside :hover

Putting transition: inside the :hover block — it won't animate back to original!

✓ **Fix:** Always put transition in the NORMAL style block.



Missing 's' after number

Writing "1" instead of "1s" — the browser ignores it!

✓ **Fix:** Always write 1s, 0.3s, 2s — never just a plain number.



Non-animatable properties

Trying to transition display or font-family — these cannot be animated.

✓ **Fix:** Stick to: color, transform, opacity, width, height, border-radius.



Property not changing

Using transition on a property that isn't actually different in :hover — nothing happens!

✓ **Fix:** Make sure the value actually CHANGES between normal and :hover.



Key Terms Recap

transition

CSS property that enables smooth change

transition: background-color 1s;

property

What to change smoothly

background-color, transform, opacity

duration

How long the change takes

0.5s (fast) / 1s (medium) / 3s (slow)

:hover

Trigger — activates on mouse hover

.box:hover { background-color: green; }

selector

Which HTML element to style

.box { ... }

transform: scale()

Resize the element

scale(1.3) makes it 30% bigger



Homework Challenge

1

MUST DO

Button with Colour + Size Change

Create a `<button>` that changes background colour AND grows (`transform: scale(1.1)`) on hover. Use transition for both. Try from memory first!

2

BONUS

Square to Circle

A square div (`border-radius: 0%`). On hover → circle (`border-radius: 50%`). Smooth transition + colour change at the same time!

3

SHOW OFF 🏆

Teach Your Family!

Show your webpage to your family. Explain CSS Transition in your own words. What does `:hover` do? Teaching = the best way to remember!

🔍 Bonus: Look up transition-timing-function — try *ease*, *ease-in*, *ease-out*, *linear*. Which do you like best?



You're a CSS Animator Now!

Every button glow, every smooth hover — you now know how it's made!

★ Stars twinkling in a night sky 🪔 Diwali diya flame flickering ⌚ A loading spinner ✨ Elements that fade in on page load

Session Takeaways

- ✓ CSS Transition makes changes smooth — no instant jumps
- ✓ transition property goes in the NORMAL style block
- ✓ :hover is the trigger — fires on mouse over
- ✓ Duration controls speed — try 0.3s, 1s, 3s
- ✓ You can transition multiple properties with commas